

Exo 1 - Hospital management System

Task 1

CLASS Patient

Attributes:

- patientID
- name
- email
- password

Methods:

- login()
- submitHealthProblem(Description)
- makePayment(amount)

CLASS Doctor

Attributes:

- doctorID
- name
- specialty
- licenseNumber

Methods:

- reviewHealthProblem(problem)
- createPrescription(problem)
- receivePayment(payment)

CLASS Organizer

Attributes:

- organiserID
- name
- email
- password

Methods:

- readHealthProblem(problem)
- consultDoctor(problem, doctor)
- sendPrescriptionToPatient(prescription, patient)
- forwardPayment(payment, doctor)
- createMember()
- issueCredentials()
- maintainDatabase()

Relationships

Patient "0..*" --- "1" Organizer

Organizer "1" --- "0..*" Doctor

Patient "0..*" --- "0..*" Doctor

Task 2

CLASS HealthProblem

Attributes:

- problemID
- description
- createdAt
- status

Methods:

- setStatus(status)
- addNote(note)

CLASS Prescription

Attributes:

- prescriptionID
- content
- createdAt
- validUntil

Methods:

- generatePrescription()
- markAsSent()

CLASS Payment

Attributes:

- paymentID
- amount
- date
- status

Methods:

- processPayment()
- refundPayment()

Payment Inheritance

Payment

- Cash Payment
- Check Payment
- Credit Card Payment

Updated Relationships

Patient "1" --- "0..*" HealthProblem

Organizer "1" --- "0..*" HealthProblem

Doctor "0..*" --- "0..*" HealthProblem

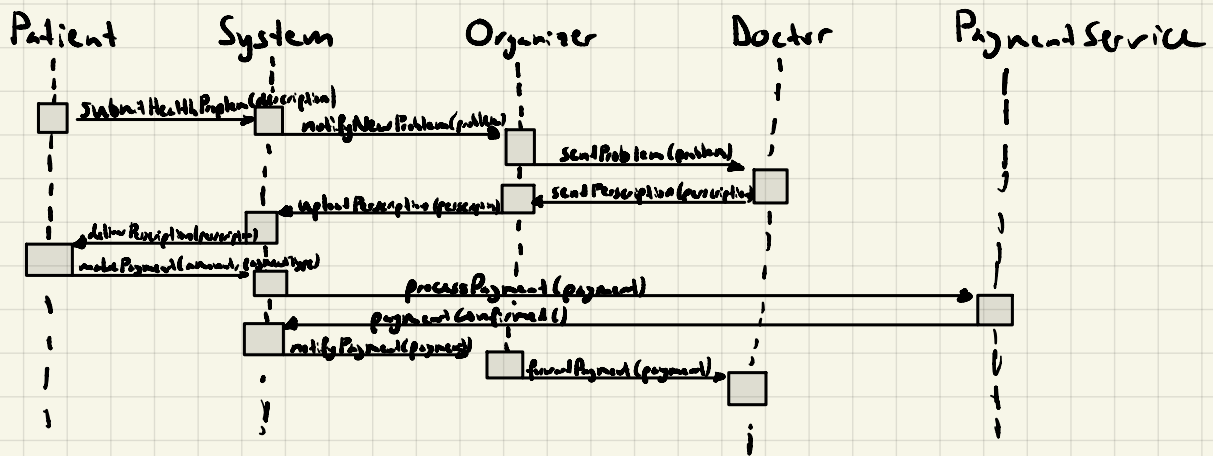
HealthProblem "1" *--- "0..1" Prescription

Patient "1" --- "0..*" Payment

Organizer "1" --- "0..*" Payment

Doctor "1" --- "0..*" Payment

Sequence Diagram



Task 3

CLASS User

Attributes:

- id
- name
- email
- password

Methods:

- login()
- logout()
- change Password()

INHERITANCE

User

- Patient
- Doctor
- Organizer

STATUS MANAGEMENT

ENUM ProblemStatus

- PENDING
- IN_REVIEW
- RESOLVED

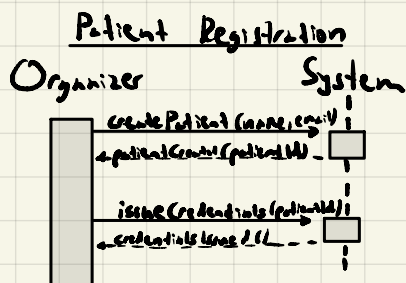
Health Problem

- status: ProblemStatus
- updated At
- resolved At

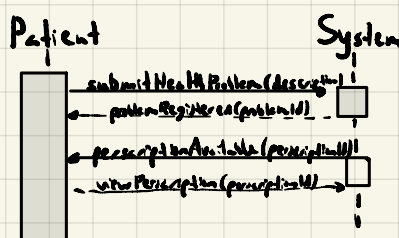
PAST PRESCRIPTIONS

Patient "1" -- "0..*" Prescription

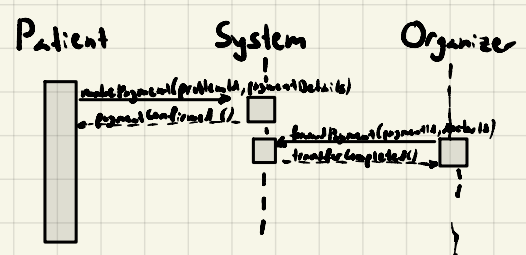
SYSTEM SEQUENCE DIAGRAMS



Consultation



Payment



Exo 2- ABCD Records

Task 1.1

ACTORS:

- Member
- Order Processing Clerk

USE CASES:

- Place Order
- Verify Membership
- Process Order
- Make Payment

LINKS:

Member → Place Order

Member → Make Payment

Order Processing Clerk → Verify Membership

Order Processing Clerk → Process Order

Task 1.2

CLASS Member

Attributes: memberId, name, address, memberType

Methods: placeOrder(), getDiscountRate(), canOrderUnavailable()

CLASS Order

Attributes: orderId, date, status, totalAmount

Methods: addLine(), calculateTotal(), markProcessed()

CLASS OrderingProcessingClerk:

Attributes: clerkId, name, email, shift

Methods: verifyMembership(), processOrder(), verifyItemAvailability()

RELATIONSHIPS

Member "1" --- "0..*" Order

Order Processing Clerk "1" --- "0..*" Order

Task 2.1

ACTORS:

- Member
- Non Member
- Order Processing Clerk
- Collection Department Clerk

USE CASES:

- Place Order
 - Verify Membership
 - Process Order
 - Make Payment
 - Verify Item Availability
 - Apply Discount
 - Print Invoice
 - Print Shipping List
 - Order Unavailable Items
 - Send Membership Application.
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INCLUDE/EXTEND

Place order <<include>> Verify Membership
Process order <<include>> Verify Item Availability
Process order <<include>> Print Invoice
Process order <<include>> Print Shipping List
Apply Discount <<include>> Process Order
Order Unavailable Items <<include>> Verify Item Availability
NonMember → Place Order <<include>> Start Membership Application

ACTOR LINES

Member → Place Order
NonMember → Place Order
Order Processing Clerk → Verify Membership
Order Processing Clerk → Process Order
Collection Department Clerk → Make Payment

Task 2.2

CLASS RegularMember (Member)

Attributes: discountRate
Methods: canOrderUnavailable()

CLASS RoyalMember (Member)

Attributes: discountRate, priorityLevel
Methods: canOrderUnavailable()

CLASS OrderLine

Attributes: quantity, unitPrice
Methods: lineTotal()

CLASS Item

Attributes: itemId, title, price, available
Methods: isAvailable()

CLASS CD (Item)

Attributes: artist
Methods: getType()

CLASS Tape (Item)

Attributes: duration
Methods: getType()

CLASS Invoice

Attributes: invoiceId, issueDate, total
Methods: printInvoice()

CLASS ShippingList

Attributes: shippingListId, shipDate
Methods: printShippingList()

CLASS Payment

Attributes: paymentId, amount, paidAt, status
Methods: process()

CLASS CashPayment (Payment)

Attributes: receiptNumber
Methods: process()

CLASS CheckPayment (Payment)

Attributes: checkNumber, bankName
Methods: process()

CLASS BankDraftPayment (Payment)

Attributes: draftNumber, bankName
Methods: process()

CLASS MembershipApplication

Attributes: applicationId, name, address, date
Methods: submit(), approve()

CLASS CollectionDepartmentClerk

Attributes: clerkId, name, email
Methods: recordPayment(), validatePayment()

RELATIONSHIPS

Member "1" -- "0..*" Order

Order "1" -- "1..*" OrderLine

OrderLine "1" -- "1" Item

Order "1" -- "0..2" Invoice

Order "1" -- "0..2" ShippingList

Order "1" -- "0..1" Payment

Member "1" -- "0..*" Payment

OrderProcessingClerk "1" -- "0..*" Order

CollectionDepartmentClerk "1" -- "0..*" Payment

NonMember "1" -- "0..1" MembershipApplication